Kernix 0.11.1 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

149 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



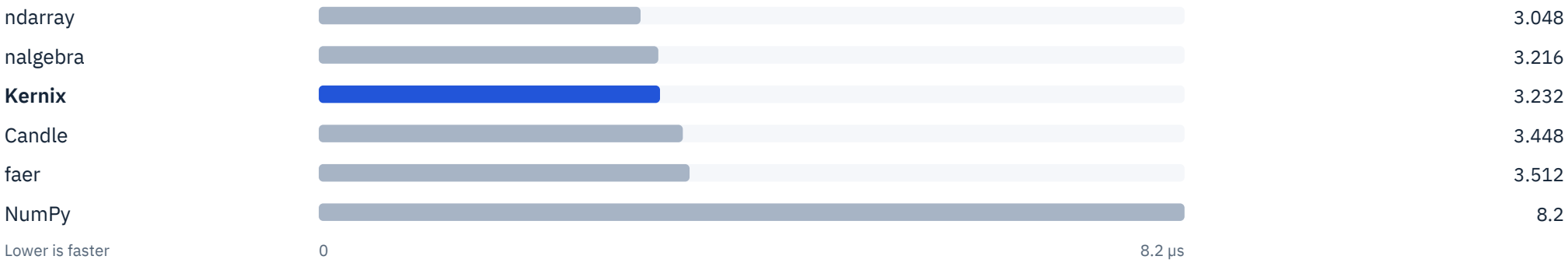
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

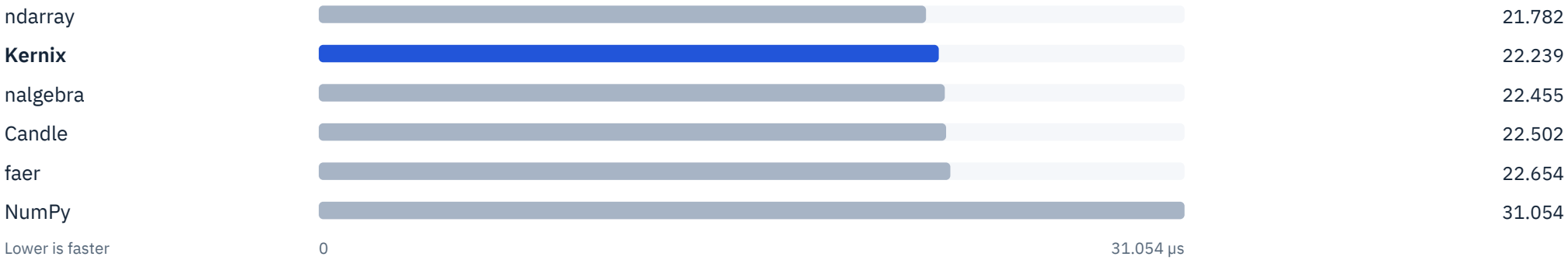
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



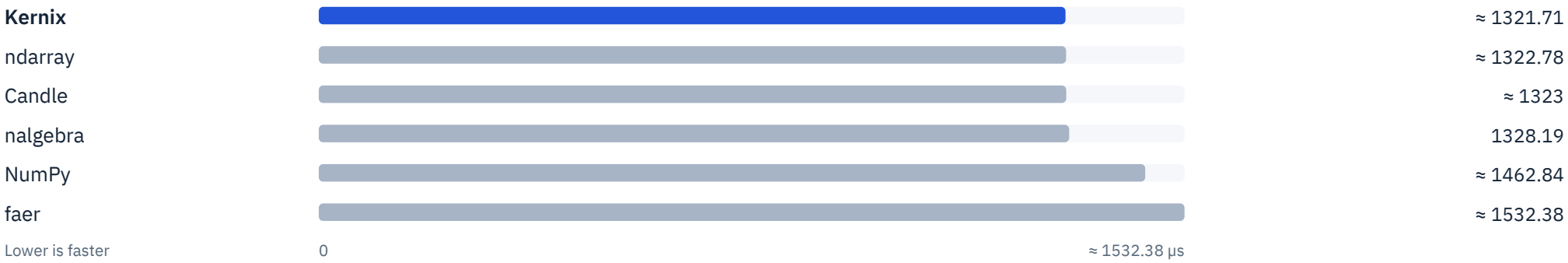
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

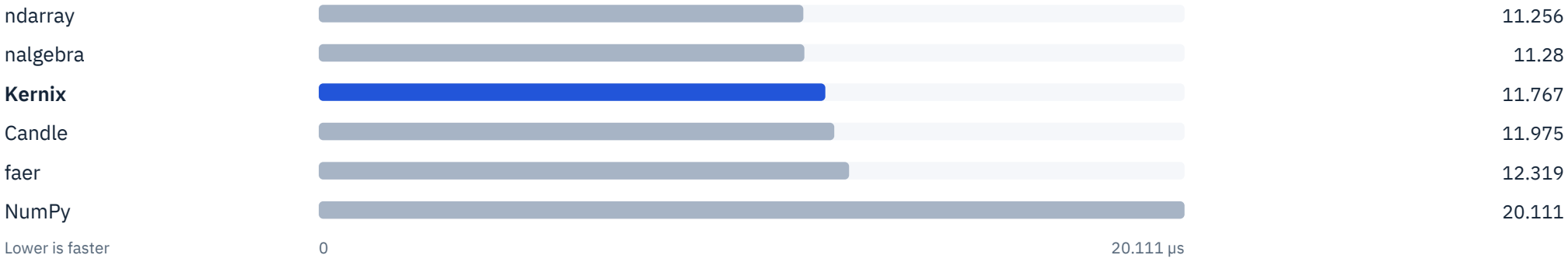
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



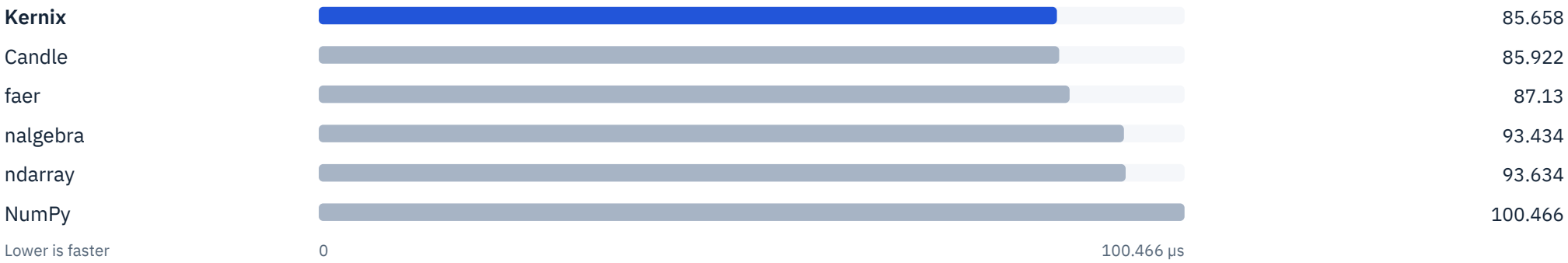
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

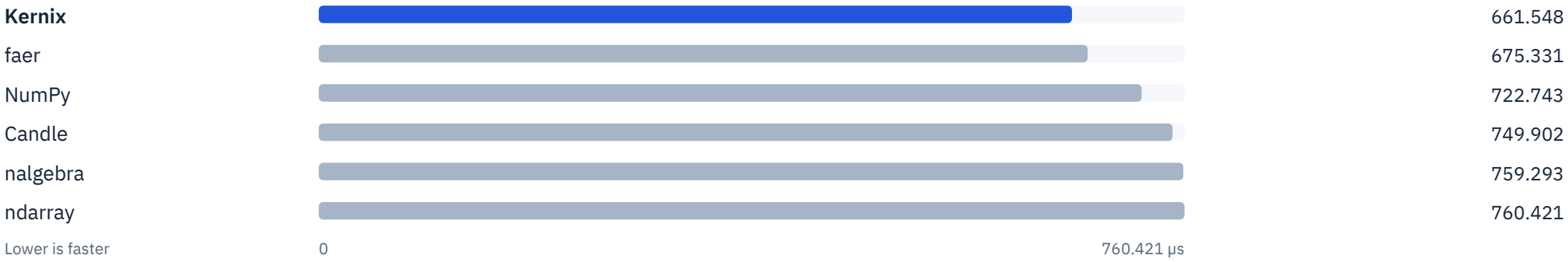
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

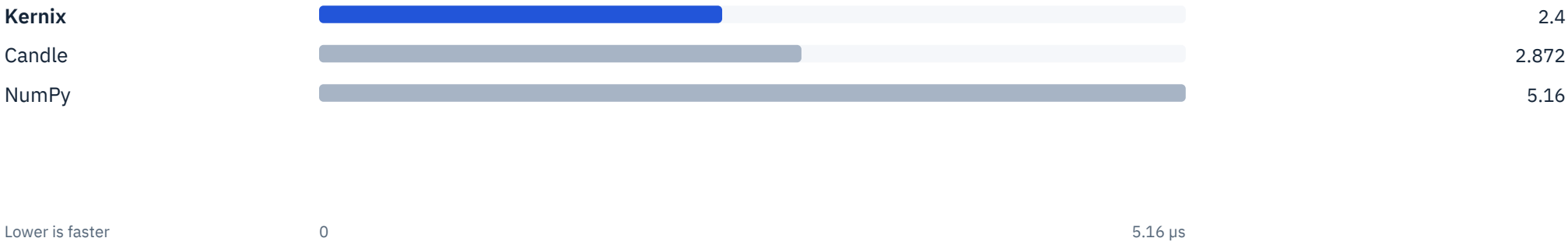
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



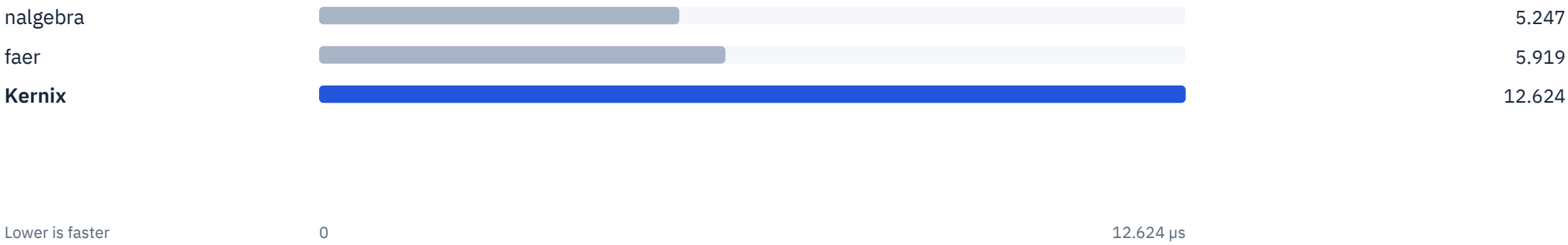
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

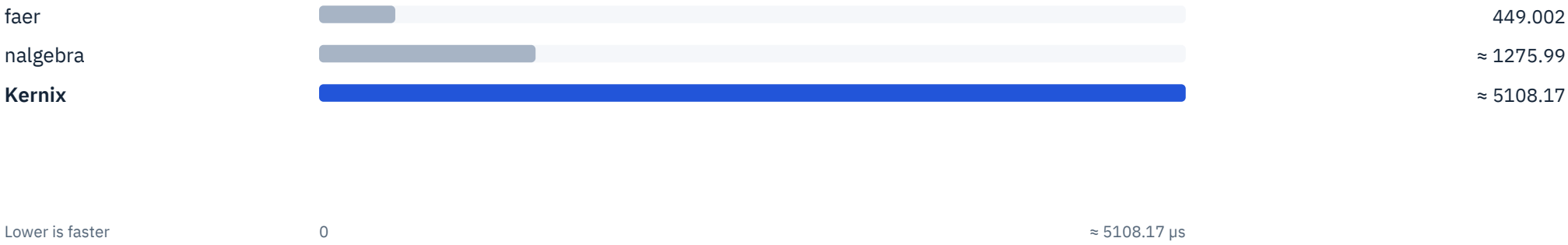
128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



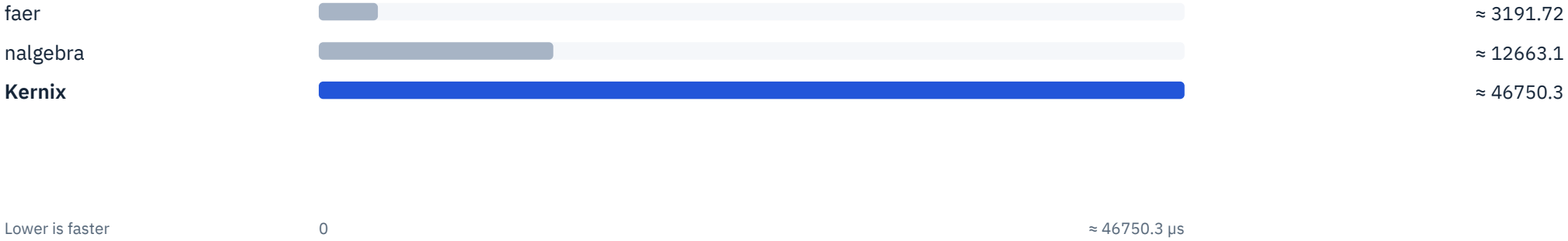
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

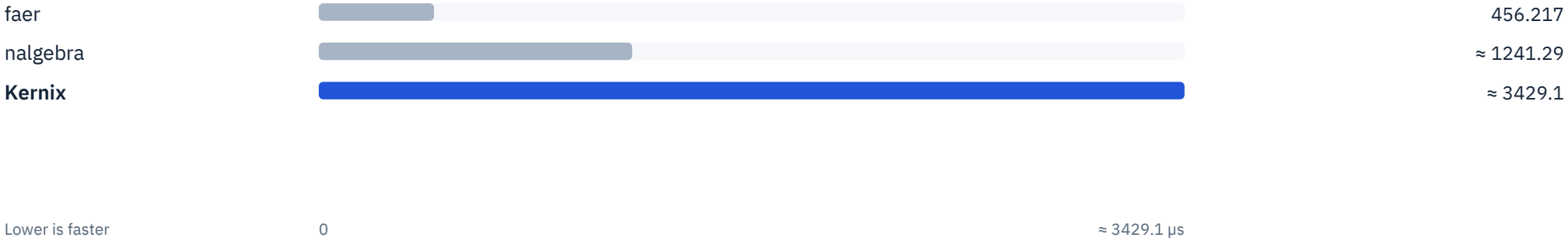
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



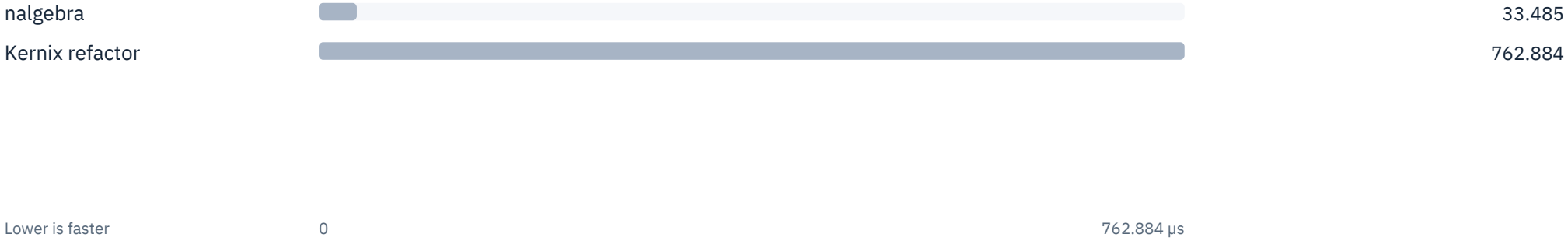
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

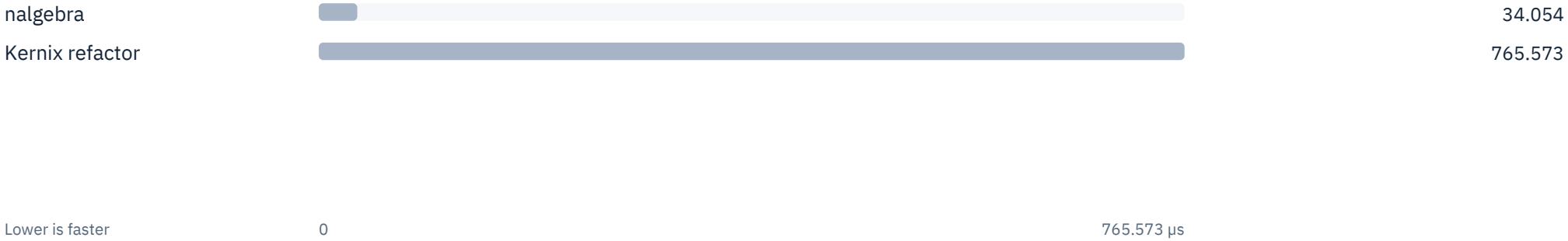
Per-call median



Cholesky rank-one downdate

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

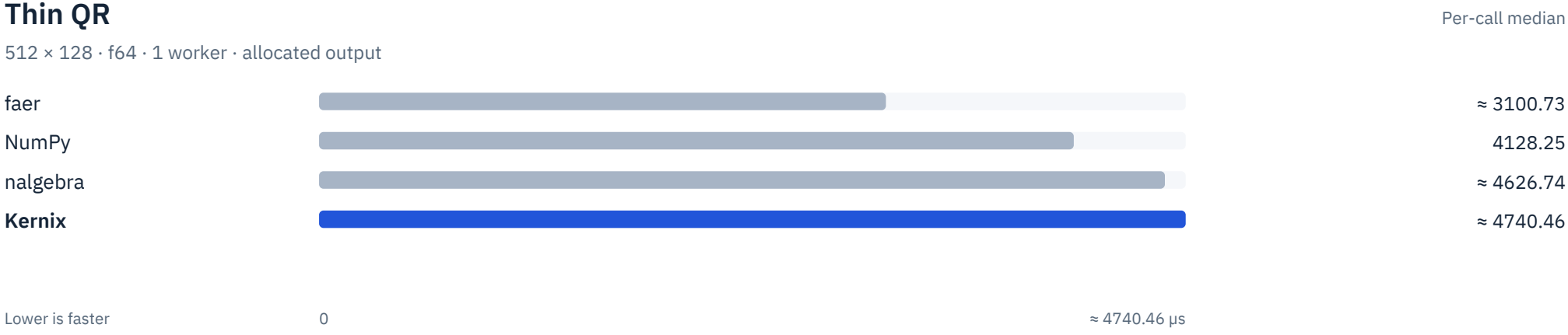
Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output



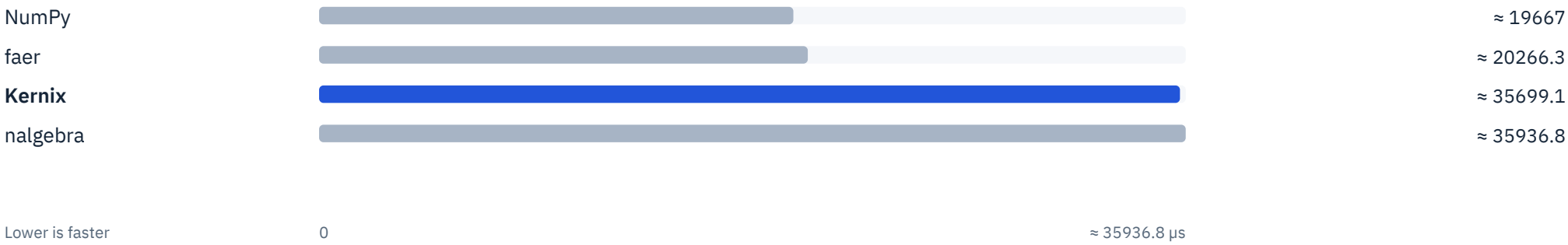
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

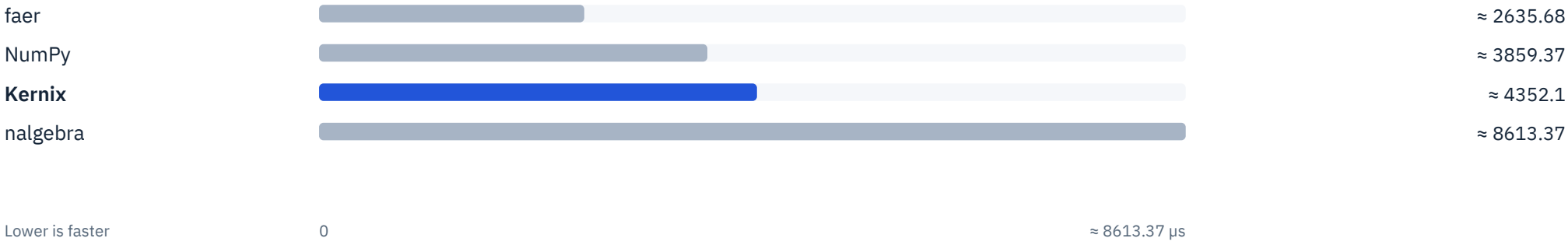
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



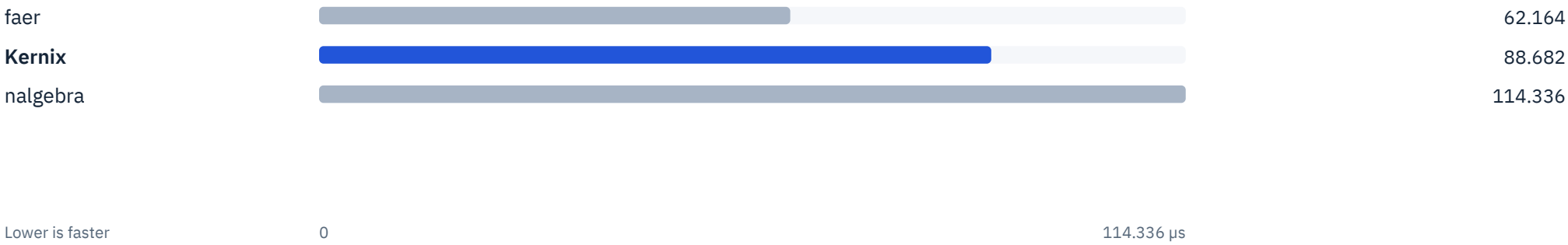
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

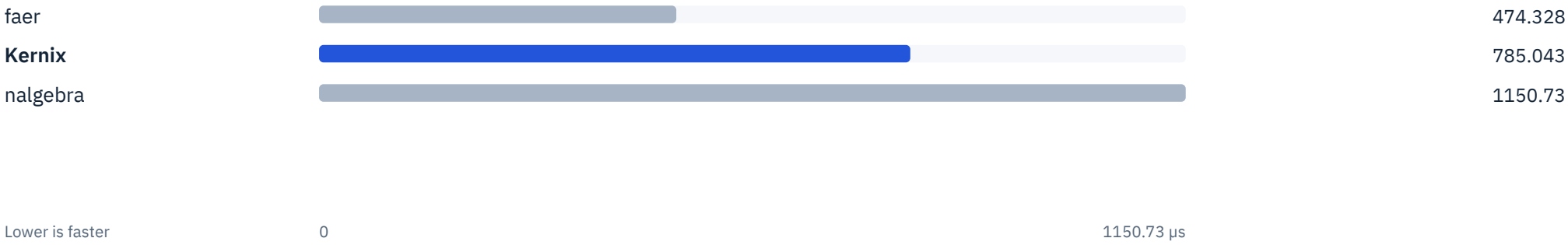
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



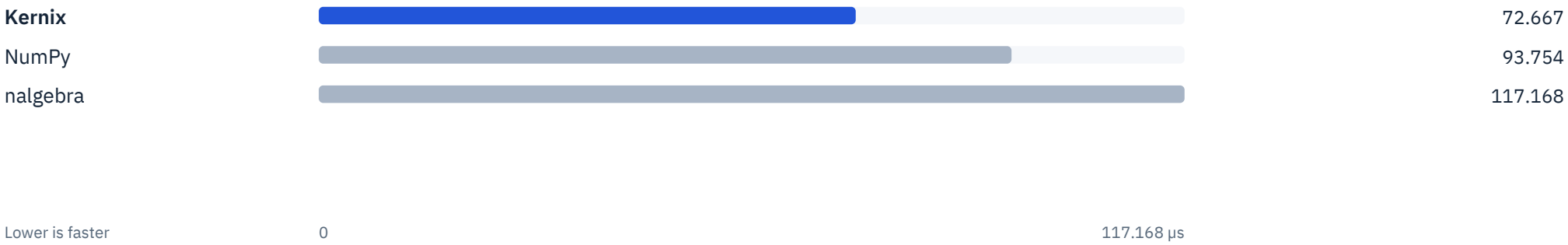
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

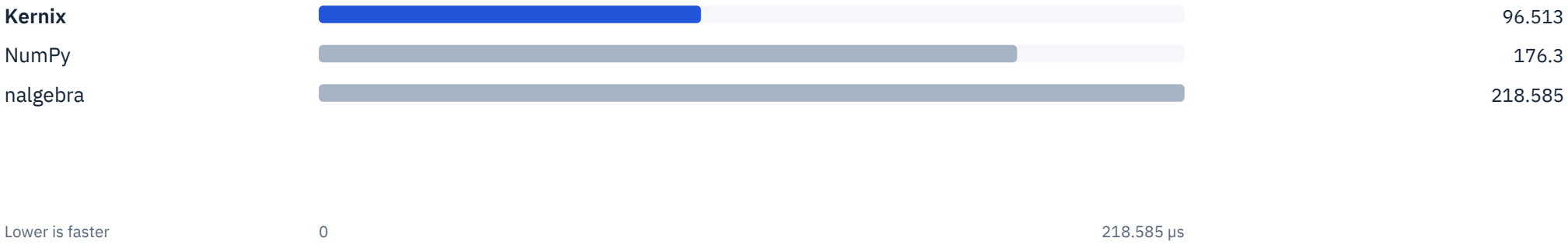
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

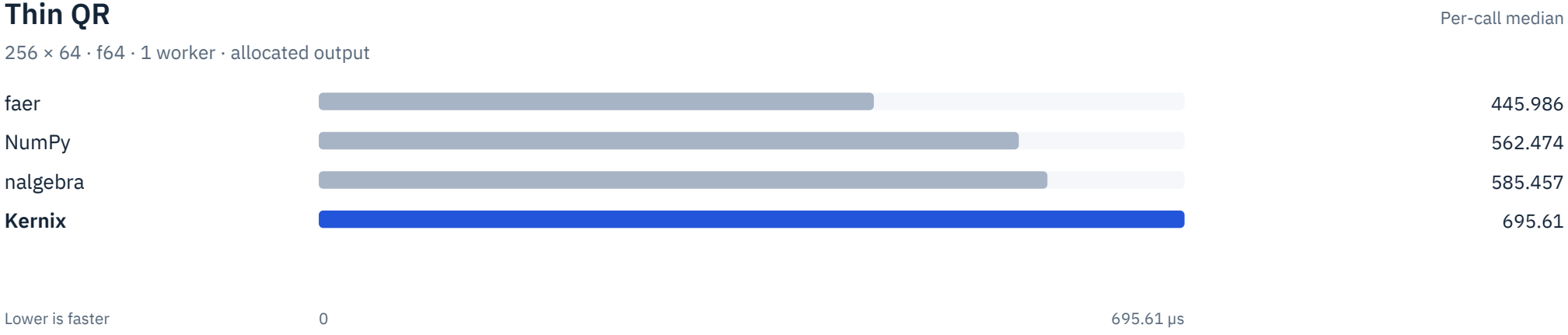
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



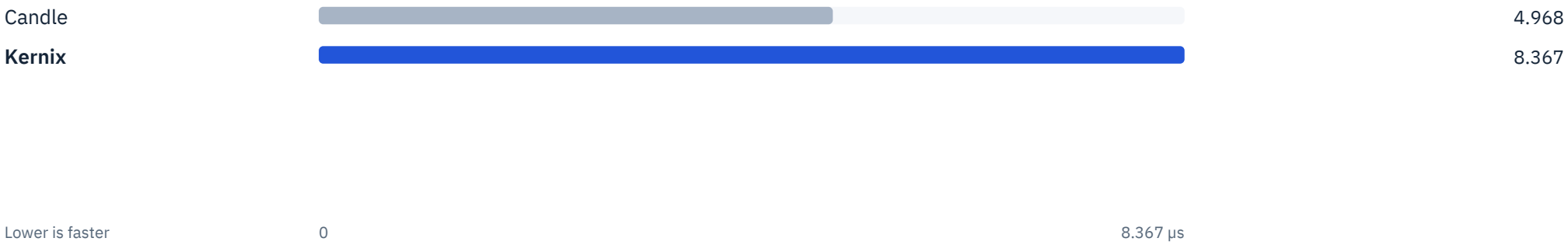
Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



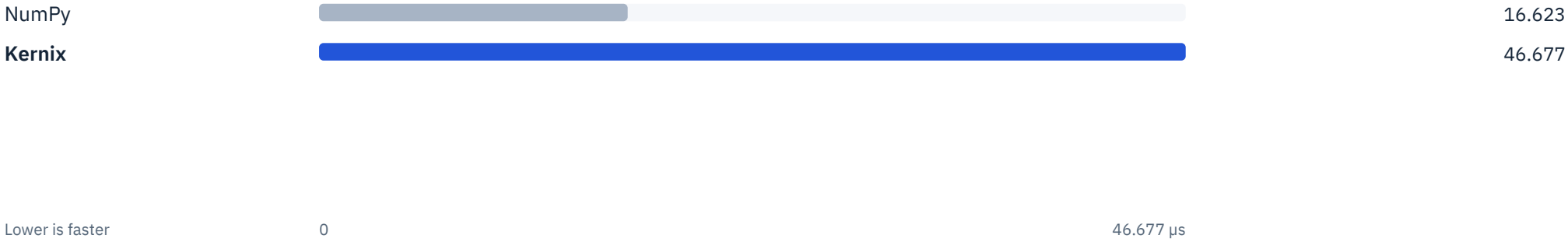
Kernix 0.11.1 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



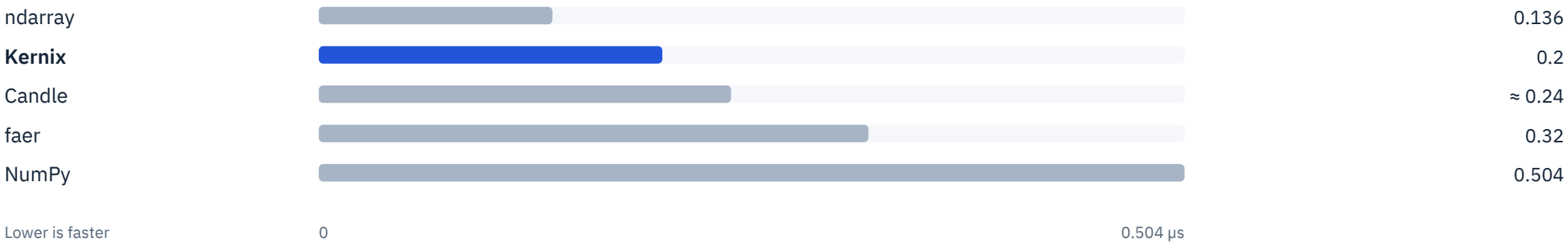
Kernix 0.11.1 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	197507	197.507
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	215162	215.162
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	100249	100.249
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	92370	92.37
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	168973	168.973
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	195667	195.667
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	85747	85.747
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	99978	99.978
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3232	3.232

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3448	3.448
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8200	8.2
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3512	3.512
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31054	31.054
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21782	21.782
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22502	22.502
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22455	22.455
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22239	22.239
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22654	22.654

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	169069	169.069
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	195170	195.17
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	210138	210.138
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168829	168.829
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169260	169.26
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	207458	207.458
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1462838	1462.838
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1328190	1328.19
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1321711	1321.711
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1322999	1322.999

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1322775	1322.775
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1532384	1532.384
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11767	11.767
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11280	11.28
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20111	20.111
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12319	12.319
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11256	11.256
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11975	11.975
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	100466	100.466
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93634	93.634

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93434	93.434
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85922	85.922
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85658	85.658
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87130	87.13
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	759293	759.293
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	661548	661.548
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	675331	675.331
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	722743	722.743
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	749902	749.902
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	760421	760.421

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12960	12.96
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3832	3.832
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4464	4.464
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5160	5.16
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2400	2.4
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2872	2.872
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16847	16.847
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	93034	93.034
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	696881	696.881
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	49061	49.061

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5919	5.919
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5247	5.247
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	12624	12.624
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	5108168	5108.168
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1275986	1275.986
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	449002	449.002
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	46750296	46750.296
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	3191721	3191.721
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	12663051	12663.051
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1241292	1241.292

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	3429105	3429.105
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	456217	456.217
Cholesky rank-one update / nalgebra 256 × 256 · f64	33485	33.485
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	762884	762.884
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	765573	765.573
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	34054	34.054
Cholesky rank-one update / nalgebra 512 × 512 · f64	146262	146.262
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4357818	4357.818
Thin QR / nalgebra 512 × 128 · f64	4626744	4626.744
Thin QR / NumPy 512 × 128 · f64	4128250	4128.25

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 512 × 128 · f64	3100728	3100.728
Thin QR / Kernix 512 × 128 · f64	4740465	4740.465
Thin QR / Kernix 1024 × 256 · f64	35699145	35699.145
Thin QR / faer 1024 × 256 · f64	20266282	20266.282
Thin QR / NumPy 1024 × 256 · f64	19667011	19667.011
Thin QR / nalgebra 1024 × 256 · f64	35936844	35936.844
Cholesky factorization / nalgebra 512 × 512 · f64	8613371	8613.371
Cholesky factorization / faer 512 × 512 · f64	2635678	2635.678
Cholesky factorization / Kernix 512 × 512 · f64	4352098	4352.098
Cholesky factorization / NumPy 512 × 512 · f64	3859372	3859.372

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 96 × 32 · f64	114336	114.336
Column-pivoted QR / faer 96 × 32 · f64	62164	62.164
Column-pivoted QR / Kernix 96 × 32 · f64	88682	88.682
Column-pivoted QR / Kernix 256 × 64 · f64	785043	785.043
Column-pivoted QR / faer 256 × 64 · f64	474328	474.328
Column-pivoted QR / nalgebra 256 × 64 · f64	1150730	1150.73
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72667	72.667
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	117168	117.168
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93754	93.754
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218585	218.585

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176300	176.3
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96513	96.513
Thin QR / faer 96 × 32 · f64	65275	65.275
Thin QR / NumPy 96 × 32 · f64	66099	66.099
Thin QR / nalgebra 96 × 32 · f64	61332	61.332
Thin QR / Kernix 96 × 32 · f64	71763	71.763
Thin QR / nalgebra 256 × 64 · f64	585457	585.457
Thin QR / Kernix 256 × 64 · f64	695610	695.61
Thin QR / faer 256 × 64 · f64	445986	445.986
Thin QR / NumPy 256 × 64 · f64	562474	562.474

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 64 × 64 · f64	18006	18.006
Cholesky factorization / Kernix 64 × 64 · f64	27326	27.326
Cholesky factorization / nalgebra 64 × 64 · f64	20343	20.343
Cholesky factorization / faer 64 × 64 · f64	13248	13.248
Cholesky factorization / NumPy 128 × 128 · f64	83691	83.691
Cholesky factorization / faer 128 × 128 · f64	65132	65.132
Cholesky factorization / Kernix 128 × 128 · f64	277221	277.221
Cholesky factorization / nalgebra 128 × 128 · f64	128879.00000000001	128.87900000000001
Cholesky factorization / nalgebra 256 × 256 · f64	1030499	1030.499
Cholesky factorization / faer 256 × 256 · f64	402901	402.901

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	749182	749.182
Cholesky factorization / Kernix 256 × 256 · f64	767732	767.732
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4968	4.968
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8367	8.367
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7343	7.343
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3512	3.512
Model inference / Kernix batched normalized/sliced MLP · f64	3032	3.032
Model inference / Candle batched normalized/sliced MLP · f64	5287	5.287
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3846.9999999999995	3.8469999999999995
Model recording + forward / Candle batched normalized/sliced MLP · f64	7608	7.608

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	4207	4.207
Model backward / Candle prepared graph → fresh gradients · f64	21086	21.086
Parameter update / Kernix transactional SGD, fixed gradients · f64	736	0.736
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16623	16.623
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46677	46.677
Row softmax / ndarray 256 × 64 · f64	97385	97.385
Row softmax / Candle 256 × 64 · f64	110345	110.345
Row softmax / Kernix 256 × 64 · f64	113144	113.144
Row softmax / NumPy 256 × 64 · f64	134623	134.623
Create zeros / Kernix 64 × 64 · f32	200	0.2

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	504	0.504
Create zeros / Candle 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / faer 64 × 64 · f32	320	0.32
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / NumPy 64 × 64 · f32	1520	1.52
Create ones / Kernix 64 × 64 · f32	248	0.248
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / ndarray 64 × 64 · f32	208	0.208